

AI 100 Final Report - Drake Davies

i) I've learned a lot about GenAI and interacting with it through code in this final project. First off, I've come to understand how GenAI can be extremely helpful and effective when acting as a tutor. I didn't always know what my changes to the code were doing and that resulted in bad self reflections. GenAI, in this case, Gemini, was very helpful at using my prompt with the issues and my reflection, and turning it back on me in the form of useful guidance while also not outright telling me. This experience has shown me how beneficial GenAI can be in coding specifically, but also as a learning enhancement in general. The GenAI is so good as a tutor because it is both informative but also supportive, such as when I was wrong, it would tell me that I was just a bit superficial, and kindly offered alternative approaches to find the right answer. When I was right, it would be encouraging and reiterate the correct answer. Overall, this project helped me understand how to interact with a GenAI and how to guide questions to get responses you want, as well as teaching me how useful GenAI can be as a partner.

ii) I learned that building an AI system is very particular and needs heavy attention to detail. It is very important to make sure each line of code you write is properly worded and necessary to the system overall. For example, in my code it was crucial for the activation function to be sigmoid, because later in the code there was a binary crossentropy loss function, and these correlated, where if there was a relu activation function, the system would not work. I also learned how important it is to get certain numbers right, such as the max length or number of words in my scenario. To me, at first, these numbers seemed relatively arbitrary, but through testing and debugging, I learned how crucial it is to get these smaller details correct. Another aspect of AI systems I discovered was the idea of training and test data. I came to realize how important it is to use these variables and the data in the right spots, like when I switched a testing variable for a training variable, and the integrity of the model was hurt. In general, I learned how the different lines of code are so important and interconnected when building an AI system.